

BALKASHINO, KazakhstanWMO: **289780**Lat: **52.5283N**Lon: **68.7428E**Elev: **1306**StdP: **14.02**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-22.2	-33.5	0.9	-27.9	-27.5	1.2	-21.7	26.3	15.9	23.9	15.8	2.9	230	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	86.6	63.1	83.3	62.5	80.0	61.7	66.9	79.4	65.5	77.9	64.0	75.9	9.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.8	89.8	71.2	61.1	84.5	69.8	59.5	79.8	68.5	32.2	79.3	31.1	77.9	29.9	76.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	21.8	18.9	-34.7	91.7	5.2	3.0	-38.4	93.9	-41.4	95.6	-44.3	97.3	-48.1	99.5		
(5)			-34.8	69.8	5.1	1.4	-38.5	70.8	-41.5	71.7	-44.4	72.5	-48.1	73.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.1	1.4	4.6	18.1	38.5	53.2	62.0	64.1	61.8	51.0	38.1	19.7	7.1		
	DBStd	25.03	12.91	12.04	11.14	11.13	8.71	7.53	6.14	7.48	8.53	9.21	13.13	12.99		
	HDD50	6923	1506	1272	988	367	69	7	0	5	88	383	908	1330		
	HDD65	11084	1971	1692	1453	795	375	141	90	153	426	835	1358	1795		
	CDD50	1494	0	0	0	22	167	367	438	369	118	13	0	0		
	CDD65	180	0	0	0	0	8	51	63	52	6	0	0	0		
	CDH74	2531	0	0	0	8	199	735	747	696	145	1	0	0		
	CDH80	703	0	0	0	0	38	211	199	215	40	0	0	0		
	WSAvg	7.8	6.7	8.2	8.7	8.5	8.7	7.6	7.0	6.6	7.0	7.6	8.3	8.3		
	PrecAvg	15.9	0.9	0.7	0.9	0.9	1.5	1.7	3.0	1.8	1.0	1.3	1.2	1.1		
(15) Precipitation	PrecMax	24.9	2.0	1.3	2.0	2.2	3.5	3.7	7.3	5.1	2.5	2.6	2.7	2.5		
	PrecMin	9.8	0.1	0.0	0.1	0.1	0.2	0.6	0.9	0.3	0.2	0.2	0.4	0.3		
	PrecStd	3.9	0.5	0.3	0.6	0.6	0.9	0.9	1.6	1.3	0.6	0.6	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.5	32.3	44.4	73.8	84.5	89.6	90.4	90.4	85.6	70.4	46.8	32.4		
		MCWB	30.9	30.7	36.9	54.1	58.4	64.1	65.0	63.8	59.3	51.0	42.0	30.8		
	2%	DB	27.9	28.9	38.4	67.4	79.3	85.4	85.2	85.7	78.0	63.8	41.6	30.4		
		MCWB	27.2	27.9	34.6	50.3	57.4	62.3	64.3	63.3	56.4	48.4	38.8	29.6		
	5%	DB	24.2	25.9	35.1	62.0	75.2	82.2	81.3	81.6	71.8	58.2	37.7	27.5		
		MCWB	23.3	24.8	32.7	48.4	55.5	62.4	63.8	62.8	54.3	46.1	35.5	26.5		
	10%	DB	20.8	22.7	32.7	56.3	71.3	78.1	78.0	77.4	66.7	52.8	34.5	24.6		
		MCWB	19.9	21.5	31.1	45.8	54.3	60.8	63.3	61.3	52.2	43.9	32.9	23.6		
	0.4%	WB	31.0	30.9	38.7	56.0	62.4	67.7	69.1	67.9	62.2	53.1	43.5	31.7		
		MCDB	31.3	32.1	42.8	69.4	74.6	81.7	79.8	81.2	80.3	65.5	45.4	32.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.3	28.0	35.0	52.1	59.2	65.6	67.4	66.1	58.4	50.1	38.8	29.6		
		MCDB	27.9	28.9	37.6	63.7	73.7	79.6	79.0	79.0	71.8	59.9	41.3	30.2		
	5%	WB	23.3	24.8	33.1	49.3	57.5	64.0	66.1	64.5	55.7	47.1	35.8	26.6		
		MCDB	24.1	25.7	35.0	60.0	71.8	77.7	77.9	77.8	69.1	56.4	37.4	27.4		
	10%	WB	19.8	21.5	31.3	46.3	55.6	62.3	64.6	62.6	53.4	44.7	32.8	23.7		
		MCDB	20.7	22.7	32.9	56.1	68.5	74.8	75.0	74.7	64.0	52.0	34.1	24.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	16.4	16.2	19.2	24.1	23.2	21.0	23.2	22.3	17.0	12.4	13.7		
		MCDBR	12.7	14.4	14.0	29.1	30.7	29.7	27.8	30.4	30.8	28.0	13.3	13.0		
		MCWBR	12.7	13.8	11.1	15.3	13.8	12.2	11.2	12.1	13.7	15.7	10.9	12.7		
	5% WB	MCDBR	12.9	13.2	11.9	25.8	26.3	25.5	23.0	25.7	27.7	24.2	12.2	12.8		
		MCWBR	13.1	13.0	10.0	15.0	13.5	11.7	10.9	11.9	13.4	14.6	10.6	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.310	0.364	0.400	0.397	0.399	0.418	0.410	0.358	0.339	0.313	0.280		
	taud		2.079	2.044	1.963	2.135	2.241	2.309	2.244	2.250	2.351	2.296	2.140	2.061		
	Ebn at Noon		215	246	254	263	271	271	263	258	259	236	200	189		
	Edh at Noon		24	34	45	43	41	39	41	38	31	26	23	20		
(44) All-Sky Solar Radiation	RadAvg		396	754	1201	1495	1831	1933	1789	1555	1133	605	287	267		
	RadStd		30	38	132	145	163	163	134	120	82	87	41	36		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page